PRACTICAL 7

A computer system has four different types of resources (printer, ROM, hard disk and RAM) and it needs to complete execution of five processes (Google Drive, Firefox, Word Processor, Excel and PowerPoint)' The number of allocated resources, maximum needed resources to complete the execution and total available resources should be taken from the user. (Sample data for reference is given below)

Process	Allocation				MAX			
	Printer	ROM	Hard Disk	RAM	Printer	ROM	Hard Disk	RAM
Google Drive	0	0	1	4	0	6	5	6
Firefox	0	6	3	2	0	6	5	2
Word Processor	0	0	1	2	0	0	1	2
Excel	1	0	0	0	1	7	5	0
PowerPoint	1	3	5	4	2	3	5	6

Available Resources			
Printer	ROM	Hard Disk	RAM
1	6	2	0

```
        {
            if(finish[i]==0)
            {
                int flag = 0;

                for(j=0;j<resources;j++)
                {
                    if(need[i][j] > available[j])
                    {
                        flag = 1;
                        break;
                    }
                }

                if(flag==0)
                {
                    for(k=0;k<resources;k++)
                        available[k] += allocation[i][k];

                    safeSequence[count++] = i;
                    finish[i] = 1;
                    found = 1;
                }
            }
        }

        if(found==0)
        {
            printf("\nSystem is in Deadlock\n");
            return 0;
        }
    }

    printf("\nSafe Sequence is:\n");

    for(i=0;i<processes;i++)
        printf("P%d ",safeSequence[i]);

    printf("\n");

    return 0;
}
```

```
(base) sakshirakhade@Sakshis-MacBook-Air ~ % cd Desktop
(base) sakshirakhade@Sakshis-MacBook-Air Desktop % touch banker.c
(base) sakshirakhade@Sakshis-MacBook-Air Desktop % nano banker.c
(base) sakshirakhade@Sakshis-MacBook-Air Desktop % gcc banker.c -o banker
(base) sakshirakhade@Sakshis-MacBook-Air Desktop % ./banker
Enter Allocation Matrix:
0 0 1 4
0 6 3 2
0 0 1 2
1 0 0 0
1 3 5 4
Enter Max Matrix:
0 6 5 6
0 6 5 2
0 0 1 2
1 7 5 0
2 3 5 6
Enter Available Resources:
1 6 2 0

Need Matrix:
0 6 4 2
0 0 2 0
0 0 0 0
0 7 5 0
1 0 0 2

Safe Sequence is:
P1 P2 P3 P4 P0
(base) sakshirakhade@Sakshis-MacBook-Air Desktop %
```